

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE CODE: ITA 06

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO1: *Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)*

Assessment Tool Weightage: 20%

QUESTIONS: 15

Total Marks: 25

Annexure A

CONCEPT MAPPING

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Short Answer Test

1. With the necessary sketch, explain all the possible blocks in the machine-learning system. Give example for showing the concept of the supervised and Un-supervised learning methods in the machine learning scheme. (5 Marks)

2. Design a machine learning system for disaster management to predict events such as floods and earthquakes using historical and real-time data. Explain the choice of model, data preprocessing, and feature selection techniques.

(a) Describe the steps involved in training, validating, and evaluating the model.

(b) Explain how the system can assist in early warning and emergency response.

(5 Marks)

3. In the context of Find-S algorithm, consider the following dataset used to determine whether a day is suitable for playing tennis based on attributes such as Outlook, Temperature, Humidity, and Wind. Apply the Find-S algorithm to determine the maximally specific hypothesis. Show the step-by-step updates of the hypothesis after each positive training example and ignore all negative examples. Finally, provide the final hypothesis. (5 Marks)

Dataset:

Example	Outlook	Temperature	Humidity	Wind	Play Tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rainy	Mild	High	Weak	Yes
5	Rainy	Cool	Normal	Weak	Yes
6	Rainy	Cool	Normal	Strong	No
7	Overcast	Mild	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No

4. In the context of the Candidate Elimination algorithm, consider a dataset used to classify whether a person will enjoy a sport based on attributes such as Weather, Temperature, Humidity, and Wind. Using the Candidate Elimination algorithm, determine the version space by computing both the specific (S) and general (G) boundaries. Show the step-by-step updates after processing each training example. (5 Marks)

Dataset:

Example	Weather	Temperature	Humidity	Wind	Enjoy Sport
1	Sunny	Warm	Normal	Weak	Yes
2	Sunny	Warm	High	Strong	Yes
3	Rainy	Cold	High	Weak	No
4	Sunny	Cold	Normal	Weak	Yes
5	Rainy	Warm	Normal	Strong	No

5. In the context of the Find-S algorithm, consider the dataset used to determine whether a customer will buy a product based on attributes such as Age, Income, Student, and Credit Rating. Apply the Find-S algorithm to determine the maximally specific hypothesis. Show step-by-step updates after each positive example and ignore negative examples. Finally, provide the final hypothesis. (5 Marks)

Dataset:

Example	Age	Income	Student	Credit Rating	Buys Product
1	Young	High	No	Fair	No
2	Young	High	No	Excellent	No
3	Middle	High	No	Fair	Yes
4	Senior	Medium	No	Fair	Yes

1. Machine Learning System - Blocks + supervised & unsupervised learning

Basic block diagram

Data collection



Data pre-processing



Feature Extraction / Selection



Training Data



ML Algorithm



Trained Model.



Prediction / Classification



Evaluation & Feedback

Explanation of blocks

1. Data collection: Collect data from sensors, data bases, websites, etc.,
2. pre-processing: Remove errors, missing values and noise; normalize data
3. Features Selection: Select important features from the data.
4. Training: Give training data to an ML algorithm to learn patterns.
5. Model: The trained model is used for prediction
6. Evaluation: Check model performance using accuracy, precision, recall, etc.,
7. Feedback: Improve the model using new data.

Supervised Learning

The model learns from labelled data..

Example:

Input → Student study hours.

Output → Pass/Fail.

Algorithms Decision Tree, Linear Regression, SVM, KNN
unsupervised learning

The model learns from unlabelled data.

Example:-

Grouping customers according to their purchasing behaviour.

Algorithm: K-Means clustering.

2) ^{mini}

ML system for Disaster Management

The system predicts floods and earthquakes using historical and real-time data.

(a) Model, preprocessing and feature selection.

historical + Real time data

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Data pre-processing

↓

feature selection

↓

ML Model

↓

prediction

↓

early warning

Data used

- Flood: rainfall, river/water level, soil moisture, temperature, humidity
- Earthquake: Seismic waves, magnitudes, ground vibration, location and depth.

Data preprocessing

- Remove duplicate and incorrect data.
- Handle missing values
- Remove noise / outliers.

Features Selection

Select the most useful features using:

- Correlation
- Information gain
- Feature importance

Model choice.

- Random Forest / XGB Boost: good for classification and prediction
- LSTM: Useful for time-dependent sensor data.
- Classification can predict Disaster / NO Disaster.
- Regression can predict severity or water level.

b) Training, validation & evaluation.

1. Divide data into training, validation & testing sets.
2. Train the ML model using historical data.
3. Use validation data to tune the model.
4. Test the final model on unseen data.

Early warning and emergency response.

When the predicted risk crosses a threshold, the system can:

- Send alerts to authorities and people.
- Identify high-risk areas.
- Help evacuation planning

3) Find s Algorithm - Play Tennis.

Attributes:

outlook, Temperature, humidity, wind.

he consider only positive examples.

Step 1: Positive Example 3.

Example 3:

Overcast, Hot, High, Weak - yes

Initial hypothesis:

$h = \{\emptyset, \emptyset, \emptyset, \emptyset\}$

After Example 3:

$h = \langle \text{Overcast}, \text{Hot}, \text{High}, \text{Weak} \rangle$

Step 2: Positive Example 4

Example 4:

Rainy, Mild, High, Weak $\rightarrow$ yes

Compare with Current hypothesis

- Outlook: overcast $\neq$ Rainy $\rightarrow$?
- Temperature: Hot $\neq$ Mild $\rightarrow$?
- Humidity: High = High $\rightarrow$ High
- Wind: Weak = Weak $\rightarrow$ Weak.

Therefore: $h = \langle ?, ?, \text{High}, \text{Weak} \rangle$

Step 3: Positive Example 5.

Example 5:

Compare:

- Outlook $\rightarrow$ already ?
- Temperature $\rightarrow$?
- Humidity: High $\neq$ Normal $\rightarrow$?
- Wind $\rightarrow$ weak.

Therefore:

$h = \langle ?, ?, ?, \text{Weak} \rangle$

Step 4: Positive Example 7.

Example 7:

Therefore: $S = \langle \text{sunny}, \text{warm}, ?, ?, ? \rangle$

$G = \langle ?, ?, ?, ?, ? \rangle$

Example 3 - Negative.

Rainy, cold, High, weak $\rightarrow$ NO

G must be specialized so that it does not cover this negative example

possible

Specializations:

$\langle \text{sunny}, ?, ?, ?, ? \rangle$ $\langle ?, ?, \text{Normal}, ? \rangle$

$\langle ?, \text{warm}, ?, ? \rangle$ $\langle ?, ?, ?, \text{strong} \rangle$

So, $G = \{ \langle \text{sunny}, ?, ?, ?, ? \rangle, \langle ?, \text{warm}, ?, ?, ? \rangle, \langle ?, ?, \text{no} \rangle$

Example 4 - Positive

sunny, cold, normal, weak $\rightarrow$ YES

S becomes:

$S = \langle \text{sunny}, ?, ?, ? \rangle$

From G , hypotheses inconsistent with this example are removed.

therefore: $G = \{ \langle \text{sunny}, ?, ?, ? \rangle, \langle ?, ?, \text{normal}, ? \rangle \}$

Example 5 - Negative.

Rainy, warm, normal, strong $\rightarrow$ NO.

The remaining G hypotheses are specialized / removed as required

Final boundaries.

$S = \langle \text{sunny}, ?, ?, ? \rangle$

$G = \langle \text{sunny}, ?, ?, ? \rangle$

Final answer: The version space converges to:

Weather = sunny, while Temperature, Humidity and Wind can have any value.

Overcast, mild, Normal, Strong $\rightarrow$ Yes

wild differs:

Weak $\neq$ Strong $\rightarrow$?

therefore:

Final hypothesis

$$h = \{?, ?, ?, ?\}$$

So according to find-s, the maximally specific hypothesis

becomes completely general.

Candidate Elimination Algorithm

Attributes:

Weather, Temperature, Humidity, Wind

Notation:

• ? = any value • 0 = no value

Example 1 - Positive

Sunny, warm, Normal, Weak - Yes.

Specific boundary:

$$S = \{\text{sunny, warm, Normal, weak}\}$$

General boundary:

$$G = \{?, ?, ?, ?\}$$

Example 2 - Positive

Sunny, warm, High, Strong $\rightarrow$ Yes.

Generalize S:

• Humidity normal $\rightarrow$?

• Wind weak $\rightarrow$?

Therefore: $S = \langle \text{sunny}, \text{warm}, ? , ? \rangle$

$G = \langle ? , ? , ? , ? \rangle$

Example 3 - Negative.

Rainy, cold, High, weak $\rightarrow$ NO

G must be specialized so that it does not cover this negative example

possible specializations:

$\langle \text{sunny}, ? , ? , ? \rangle$ $\langle ? , ? , \text{Normal}, ? \rangle$

$\langle ? , \text{warm}, ? , ? \rangle$ $\langle ? , ? , ? , \text{strong} \rangle$

So, $G = \{ \langle \text{sunny}, ? , ? , ? \rangle, \langle ? , \text{warm}, ? , ? \rangle, \langle ? , ? , \text{no} \rangle$

Example 4 - Positive

Sunny, Cold, Normal, weak $\rightarrow$ YES

S becomes: $S = \langle \text{sunny}, ? , ? , ? \rangle$

From G, hypotheses inconsistent with this example are removed.

therefore: $G = \{ \langle \text{sunny}, ? , ? , ? \rangle, \langle ? , ? , \text{Normal}, ? \rangle \}$

Example 5 - Negative.

Rainy, warm, Normal, strong $\rightarrow$ NO.

The remaining G hypotheses are specialized / removed as required

Final boundaries.

$S = \langle \text{sunny}, ? , ? , ? \rangle$

$G = \langle \text{sunny}, ? , ? , ? \rangle$

Final answer: The version space converges to:

Weather = sunny, while Temperature, Humidity and Wind can have any value.

5. Find - S Algorithm - Boys product

Attributes:

Age, Income, Student, Credit Rating
only positive examples are considered.

Example 3 - positive

Middle, High, No, Fair $\rightarrow$ Yes

Initial:

$$h = \langle \phi, \phi, \phi, \phi \rangle$$

After Example 3:

$$h = \langle \text{Middle}, \text{High}, \text{No}, \text{Fair} \rangle$$

Example 4 - positive

Senior, Medium, No Fair $\rightarrow$ Yes

Compare:

Attribute	Example 3	Example 4	New hypothesis
Age	Middle	Senior	?
Income	High	Medium	?
Student	No	No	No
Credit Rating	Fair	Fair	Fair

Therefore:

$$h = \langle ?, ?, \text{No}, \text{Fair} \rangle$$

Final hypothesis

$\langle ?, ?, no, fair \rangle$

Meaning: A Customer is predicted to buy the products
when student = no and credit Rating = fair, regardless
of Age and income.